

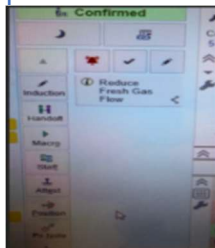
- Dr. Stone does not want us reading from slides and is okay with 3x5 notecards for presentations
- Team Discussion (**team leader** only):
 - Discussed findings with the team on 9/16
 - Texted Rachel later that day to make sure she got an interview with her stakeholder as early as possible

Weekly Meeting

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy
- Date and Time: 9/16 6:30pm-8:30pm
- Format: In-person
- Individual notes:
 - Stayed longer at the meeting to talk with Sarah about how we want to split up the design and COMSOL work
 - Finish DH01 over the weekend, so team can start reviewing
- Team Discussion (**team leader** only):
 - We used this time to work on our DH01 and make sure everyone understood the sections that were required of each other.
 - We also discussed a timeline for our solution and came up with a loose idea of what we all wanted to create.

Meeting with Interviewee

- Attendees: Brooke Struyk, Brian Struyk
- Date and Time: 9/30 6:30pm-7:30pm
- Format: Zoom
- Individual notes:
 - At CHOP and globally, anesthesiologists are talking about reducing the environmental effects of AWG
 - Low-flow is the recommended way the anesthesiologist can reduce the amount of AWG entering the air
 - When there is high-flow after a period of 15 minutes, the anesthesia machine will send a message on the notification bar. This bar is checked regularly throughout surgery. A picture of what this looks like is below.



- There has also been a push to use sevoflurane over Nitrous oxide; however, in low resource countries nitrous oxide is the most common anesthetic because it is cheap.
- Anesthesiologists typically ensure prior to the surgery that the WAGD system has been turned on and it stays on the entire day.
- Anesthesiology technicians are the main maintenance team for leaks or issues. If the issue is bigger, BMET would be brought in.
- The final say on a new technology goes more towards the BMET, and maintenance department is they can account for the new machine.



- The picture above shows the WAGD device that is being used at CHOP. The white box is a CO2 scrubber, and the flapper valve on the right has a max and min dial to show if there is enough suction for the AWG.
- Team Discussion (**team leader** only):
 - During weekly team meeting on 9/23, I was able to talk about what I learned from my stakeholder.
 - I was also able to provide insight on how pediatric anesthesiologists interact with AWG systems.

Weekly Meeting

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy
- Date and Time: 9/23 6:30pm-7:30pm
- Format: In-person
- Individual notes:
 - Complete SRR Slide by Friday
 - Prepare for meeting tomorrow with KidsOR and write down any extra questions I may come up with
- Team Discussion (**team leader** only):
 - The team created a list of questions to ask Courtney from KidsOR
 - The team created a meeting time for the SRR video recording over Zoom (decided on Friday)
 - We also made an extra meeting time after the SRR recording due date changed to Sunday. If we needed to re-record, we would be on Sunday at noon (we did not end up needing this meeting).

Meet with KidsOR Team

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy, Courtney Ludick (KidsOR), Andy Muelenaer, Dr. Robert Stone
- Date and Time: 9/24 3:30pm-4:30pm
- Format: Zoom
- Individual notes:
 - Fix SRR slides based on the meeting
 - Fix Job Map
 - Start SolidWorks tutorials because it looks like there will be more of a need
 - Read over manuals and floor plans sent to us within the next week
 - Read over full page of meeting notes written by Sarah on the Teams site
- Team Discussion (**team leader** only):
 - We were able to get most of our questions answered by Courtney
 - Dr. Muelenaer also asked interesting questions that were helpful for clarity
 - The team realized we had a misunderstanding of what our solution was and what KidsOR actually does.
 - The meeting was very helpful in clearing out confusion after DH01 research
 - The team was able to start coming up with solutions and brainstorming what direction we could go in

SRR Video Recording Meeting

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy
- Date and Time: 9/25 5:30pm-6:30pm
- Format: Teams
- Individual notes:
 - Download the video from teams
 - Edit the video and ensure it does not need to be re-recorded.
 - Submit the final video
- Team Discussion (**team leader** only):
 - We recorded in the video and prepared our slides beforehand, so the meeting was straightforward
 - We discussed future directions with our solution and what Courtney had talked about in the meeting from the day prior
 - We also discussed what we changed in our slides based on the new deliverables

Weekly Team Meeting

- Attendees: Brooke Struyk, Kaayla Osborne, Rachel Mensah, Sarah Krishnamoorthy
- Date and Time: 9/30 6:30pm-7:30pm
- Format: Zoom
- Individual notes:
 - Complete the ethics section of DH01
 - Continue to work on SolidWorks tutorials
 - Do extra research on active scavenging systems
- Team Discussion (**team leader** only):
 - Split up DH02
 - Signed the document for the DH02
 - Discussed further what our solution could be and how to prepare for our meeting with Dr. Stone on 10/1.

Literature Review

<It is expected that each team member will read and summarize **three new pieces of literature with each new submission!** Keep track of the total number below by incrementing the header. You can choose from peer-reviewed, scientific articles (<https://scholar.google.com/>), patents (<https://patents.google.com/>), engineering standards (request from Dr. Arena ahead of time if not accessible through VT or ANSI), or FDA guidance documents (<https://www.fda.gov/regulatory-information/search-fda-guidance-documents#guidancesearch>). **Do NOT use literature that was separately assigned for homework.**

Note: AI should serve as a collaborator, not a replacement, for your own critical thought and skills. The intellectual contribution and design decisions must remain your own. AI tools may lack direct access to Google Scholar/Patents and the most recent literature. Additionally, AI-generated content can be inaccurate, biased, or incomplete. Therefore, it is up to you to identify the key pieces of literature for your project. AI can be used to support the Summary of Key Findings section, but you should verify the accuracy of the content and that nothing from a piece of literature was overlooked. Please cite the AI tool used, the prompt that you provided, and your analysis of, or updates to, the content. The other sections should be completed without the use of AI.>

1. Anesthesia and its environmental impact: approaches to minimize exposure to anesthetic gases and reduce waste [1]

Source Authors, Journal, Date, Affiliation

- [1] K. Samad, Muhammad Saad Yousuf, H. Ullah, Syed Shabbir Ahmed, Khalid Maudood Siddiqui, and A. Latif, "Anesthesia and Its Environmental Impact: Approaches to Minimize Exposure to Anesthetic Gases and Reduce Waste," *Medical Gas Research*, vol. 15, no. 1, Jun. 2024, doi: 10.4103/mgr.medgasres-d-23-00059.

Summary of Key Findings

Anesthesia is a mix of nitrous oxides and halogenated ethers that are strong greenhouse gases and contribute to global warming. Operating rooms account for 20-30% of waste leaving the hospital. By improving anesthetic waste gas sensors and collecting systems or changing the flow rate of anesthetic gas, there would be less output. It is important to consider the environmental effect of anesthesia, its contribution to greenhouse gas emissions and global warming.

Application to Design Project

In our initial stakeholder meeting we talked about creating a system that would flow anesthetic gas out of a window. Our stakeholder was unsure what this would mean for the environment if this ended up as a solution, so I decided to use one of my articles to read more about it. The article talks about low-flow anesthesia, so it would help reduce the number of halogenated ethers entering the environment. It would be a good question to ask KidsOR if they are using low-flow anesthesia methods. As of right now the team also uses a chemical bed that collects this fluorine that's bad for the air and saturates it by forming mineral salts, which might be better for the environment already. This is something we can do further research on.

Specific Information for the Design project

- Acute care settings account for 8% of greenhouse gas emissions
- United States healthcare sector contributes to nearly 10% of the total GHG emissions with hospitals being the primary source
- Operating rooms generate 20-30% of the entire waste output of a hospital
- Recommended ways to minimize anesthetic gas waste exposure: LFA, advancing scavenging system, vaporizer, closed-loop delivery system, routine maintenance and calibration, and waste gas monitoring

2. An Initial Evaluation of a Novel Anesthetic Scavenging Interface [2]

Source Authors, Journal, Date, Affiliation

- [2] J. A. Barwise, L. J. Lancaster, D. Michaels, J. E. Pope, and J. M. Berry, "An Initial Evaluation of a Novel Anesthetic Scavenging Interface," *Anesthesia & Analgesia*, vol. 113, no. 5, pp. 1064–1067, Nov. 2011, doi: 10.1213/ane.0b013e31822c9a2c.

Summary of Key Findings

The primary method used in the United States and Europe is an active gas scavenging system (AGSS). The machine requires a 24-hour per day period which leads to higher energy costs. By using a DGSS,

you do not need a pump as big of a pump, energy costs are lower, and the device can be turned on and off. The article does note that there is no sensor display or failure modes that have been identified.

Application to Design Project

It was brought up during the stakeholder meeting that we should investigate anesthetic scavenging devices as a potential solution, so I decided to read more about it. The article depicts a solution from Vanderbilt University in 2011. If we were to use/ create our own scavenging system this design approach would be a good place to start. The researchers were able to save \$28,000 after a 6-month testing period. Though it would also be \$2738 per year on electricity. We need to figure out what KidsOR is using and how much it is costing them per year, so we could potentially use a similar solution and optimize it. The device built does not have any display or failure mode, so it's not necessarily what we are doing unless we wanted to start from scratch. This idea might also end up costing KidsOR more money than building onto their existing solution of the bed chamber.

Specific Information for the Design project

- Waste anesthetic gas scavenging technology has not changed much in the past 30 years
- Process requires a large vacuum pump that is expensive and operates continuously
- Standards for waste anesthetic gas were recommended but never implemented
- AGSS works by: "Exhausted gas from the anesthesia machine is removed by an open manifold connected to a vacuum source. The system entrains and evacuates waste gas with relief from positive and negative pressure in the breathing circuit. The balance is achieved either with relief valves that open to atmosphere with a slight pressure gradient or valveless 'tube-within-a-tube' system"
- Many hospitals and ambulatory centers used a dedicated scavenging system that has a separate pump and piping.

3. Waste anesthetic gas exposure and strategies for solution [3]

Source Authors, Journal, Date, Affiliation

- [3] H.-B. Deng, F.-X. Li, Y.-H. Cai, and S.-Y. Xu, "Waste Anesthetic Gas Exposure and Strategies for Solution," *Journal of Anesthesia*, vol. 32, no. 2, pp. 269–282, Feb. 2018, doi: 10.1007/s00540-018-2448-1.

Summary of Key Findings

The article brought up that different countries have different standards for how to treat anesthetic waste gas. It also brought up that long term exposure can lead to chronic diseases that based on their research harm the anesthesiologist later on. This is especially concerning because there is no standard biomarker test for elongated waste gas exposure. The article recommends using a scavenging system,, air-conditioning system, Iso-Gard Mask, or passive scavenging system with activated charcoal canister to effectively get rid of gas waste.

Application to Design Project

This article brought up a few interesting points that there is no set limit for the amount of sevoflurane given to a patient, and there is also no universal standard for waste anesthetic gas control. KidsOR travels to different countries and had to abide by different standards, so I think this would be interesting

to ask if there is some initial research they have to do before going to a new country and administering anesthesia. The article also brought up concerning biomarkers for WAG exposure after a long period of time. I think it would be interesting to ask the team if they recognize they do any yearly testing for these biomarkers. It would also be interesting to ask my stakeholder, who is a pediatric anesthesiologist, if the hospital he works at has yearly tests.

Specific Information for the Design project

- It's impossible to fully eliminate waste anesthetic gas (WAG) from the air
- Long term exposure to WAG can lead to increased oxidative stress DNA damage, genotoxicity, and carcinogenesis
 - Some have found an increase in chronic diseases, spontaneous abortion, congenital malformations, low birth rates, and higher stillbirth in offspring
- Check out the National Institute of Occupational Health and Safety and American Conference of Governmental Industrial Hygienists for future literature reviews on WAG exposure
- 28% of OR's complied with predetermined standards about WAG
- Different countries have their own limit standards (ADD ABOVE)
 - There is a table on page 3 that would be useful to reference exposure limits in different countries
- There is no set limit for the amount of sevoflurane used
- WAG is higher when patient is in a sitting position compared to standing
- Anesthesiologist is the one getting the most amount of WAG exposure compared to the rest of the OR
- There is no "good" biomarker testing WAG exposure levels

4. A Novel Approach to Scavenging Anesthetic Gases in Rodent Surgery [4]

Source Authors, Journal, Date, Affiliation

[4] J. C. Nesbitt, D. A. Krageschmidt, and M. C. Blanco, "A Novel Approach to Scavenging Anesthetic Gases in Rodent Surgery," *Journal of Occupational and Environmental Hygiene*, vol. 10, no. 9, pp. D125-D131, Sep. 2013, doi: [10.1080/15459624.2013](https://doi.org/10.1080/15459624.2013).

Summary of Key Findings

When giving laboratory animals, specifically rats, anesthetic gases, in this case isoflurane, researchers noticed excessive concentrations of the gas throughout the surgical site. When using the passive scavenging system, there five times the regular amount of waste gas in the air. The researchers came up with a nose cylinder and an exhausted induction chamber as their solutions. The solutions ended up significantly reducing the amount of isoflurane and oxygen levels to safe levels.

Application to Design Project

One of our initial deliverables was to look at how AWG is treated in veterinary settings. This article brought up two interesting solutions. Instead of making a new scavenging system, they went to the root cause and improved the mask to specifically fit rat's noses. By also creating a cylinder around the rat's full head as well, there was no significant/ unsafe amount of gas allowed to escape. The second solution was interesting because it placed the rat in a box and closed the animal in. Once inside, anesthesia was used and scavenged, and then the box was opened. Both solutions are interesting because it looks at the root cause of the issue, that is AWG and solves it. In real-life operating rooms

neither of these solutions could work, but it does highlight the importance of an active scavenging system that filters the AWG from the air.

Specific Information for the Design project

- On the chamber the minimum vacuum flow was 17 liters per minute
 - Vacuum evacuated oxygen and isoflurane
- The article brings up excessive amounts of oxygen leading to issues from the AWG, so I will investigate this when we are deciding what specific part of the AWG we want to target
- Potentially further look into anesthesia masks
 - Realistically nothing will be changed about this because it has to be a universal device for all children and sometimes adults who are different sizes, but it is important to see why the AWG escapes
- Some scavenging systems use exhaust rates of 12-15 liters/minute

5. Uptake and biotransformation of sevoflurane in humans: A comparative study of sevoflurane with halothane, enflurane, and isoflurane [5]

Source Authors, Journal, Date, Affiliation

[5] Y. Shiraishi and K. Ikeda, "Uptake and Biotransformation of Sevoflurane in Humans: A Comparative Study of Sevoflurane With Halothane, Enflurane, and Isoflurane," *Journal of Clinical Anesthesia*, vol. 2, no. 6, pp. 381–386, Nov. 1990, doi: [10.1016/0952-8180\(90\)90024-W](https://doi.org/10.1016/0952-8180(90)90024-W).

Summary of Key Findings

The article compared 4 different anesthetic gases: sevoflurane, halothane, enflurane, and isoflurane. They tested these gases for an hour on 32 patients who were healthy and monitored the amount that was absorbed by the body, release of fluoride into the blood, and the potential damages on the kidney and liver. They found that sevoflurane raised fluoride levels in the blood the most and had low absorption rates. They found that isoflurane and enflurane was broken down by the body the most. There were also no effects on the kidney or liver by any of the four gases.

Application to Design Project

Before designing our scavenging system, Dr. Stone told us it would be easiest to choose one anesthetic waste gas to be collected by the system. This way we can collect clear data on a specific gas and test for reduction in the air. The team is leaning towards sevoflurane gases because it was brought up by Courtney that this gas runs out the quickest in the hospitals. By targeting this gas, there is a potential to make a big difference in the amount of AWG left in the air. I read this article because I thought it would be important to see the differences in the anesthetic gases being used. It was also interesting to see how the body reacts to different gases used in anesthesia.

Specific Information for the Design project

- If a low amount of sevoflurane is being absorbed by the body, there is a potential that a lot would be breathed back out and contributing to the anesthetic waste gas issue

- It was brought up that the gas used up the quickest is the sevoflurane. This may be due to the low rate of absorption within the body which causes doctors to use excess amounts/ more of it.
 - It would be interesting to further look into why sevoflurane is a popular to use as anesthetic gas even with its low absorption levels.
 - Also look into the other 3 gases and how often they are used in hospital settings
- Sevoflurane required using a higher concentration when comparing it to the other gases, so they would all measure up to 1.1 minimum alveolar concentration (MAC) (the standard the researchers used).

6. Scavenging, Medical Gas Pipelines, and Vacuum Systems [6]

Source Authors, Journal, Date, Affiliation

[6] N. A. Shallik, W. A. Eldegowa, M. O. Zekri, and M. F. Alhammad, "Scavenging, Medical Gas Pipelines, and Vacuum Systems," *Improving Anesthesia Technical Staff's Skills*, pp. 137–150, 2022, doi: [10.1007/978-3-030-88849-7_10](https://doi.org/10.1007/978-3-030-88849-7_10).

Summary of Key Findings

The article walked one through each part of a waste gas scavenging system. There are four main parts to a scavenging system: collecting tubes, transferring tubes, a scavenging interface, and disposal tubing. In collecting tubes there is an adjustable pressure limiting valve (APL) and a pressure relief valve that collects the extra anesthetic gases. The transferring tubes have a scavenging outlet at the back of the anesthesia machine. The scavenging interface protects the breathing circuit from any positive or negative pressure changes. Finally, the disposal tubing carries the gas from the interface to the disposal. This is the main difference between active and passive systems.

Application to Design Project

Our main deliverable in the project is to create a standalone active scavenging system, so I thought it would be interesting to read more about the parts of these systems and how it works mechanically. The article walks one through how passive versus active scavenging systems work. It also highlights the importance of active systems and why they are more popularized in hospital settings. It also brings up a potential maintenance schedule for the scavenging machines that we could use and compare to the actual schedule KidsOR has created. The article also brings up why active systems are centrally used throughout the hospital, so by understanding this, we can create a standalone system.

Specific Information for the Design project

- The article recommends the scavenging system of the breathing system should be checked daily.
 - Documented maintenance schedules for the ventilation system in the OR, post-anesthesia care units, and the anesthesia machines
 - Depending on what Courtney sends we could additional information about this, especially for the ventilation PMs that are being used in places with passive systems
- The article recommends the scavenging system of the breathing system should be checked daily.
- Due to the fact medical gases are used in all areas of the hospital, there is a central gases control panel.

- Our solution needs to have a warning and alarm system to ensure that there is no issue in the breathing circuit and AWG levels are not getting too high.

Individual Design Updates

<This section is your electronic design notebook. Include sketches, photos, engineering drawings, equations, calculations, tables, graphs, observations, etc. **Focus on your narration of the design process. For example, your discussion of trainings that you complete and the outputs and/or challenges of that training process (e.g., COMSOL tutorial results, CAD training results, etc.).** Be sure to describe each item in detail with explanatory text so that it is easy to understand when reviewing. It is up to you on how to effectively organize the section headers. Include any notes about updates to User Needs (UN), Design Inputs (DI), Design Outputs (DO), Design Verification (VER), and Design Validation (VAL). **There should be at least 3 new pages with each new submission.** In select cases, you may paste and circle **individual items** from your Mural or Miro team boards and describe your contribution in the text below. However, most of the content in this section should be your **individual work**. Make sure that the font size is large enough to be legible, including within figures. However, **a majority of the 3 pages (≥ 1.5 pages) should be your own descriptive text or narration of the design process. Additionally, it is expected that a majority of the content ($\geq 1/2$) is new or expands beyond what was completed as part of other assignments, such as reading reflections and DHF's. If you include items from a prior individual assignment, please note that in bold/red and expand your discussion.**>

Defining User Needs and Purpose

Questions for 09/10 meeting: - These questions will have to be asked to Scotland team...

1. Talk us through the surgery process.
2. Tell us about the waste management bed and how you interact with it.
3. What specific issues are you facing with the current solution?
4. What types of surgeries typically experience the drowsiness?
5. What type of anaesthetic gas is being used?
6. Are there any other chemicals we should consider?
7. At what point in the surgical process do you realize you are drowsy?
8. What are the parameters that control the flow rate of gas?
9. send pics...
10. What do the rooms for kids OR look like?
11. What models of anaesthetic machines do you use?
12. Is there a preventative maintenance procedure for this?
13. Financing about the device? - Ask about a realistic investment in this device

Project pitch:
Imagine you're in surgery and the surgeon falls asleep in the operating room with you. When you think about nurses, doctors, and care technicians you think of long hours, night shifts, and constant lack of sleep. On top of already exhausted medical personnel, anesthetic gas may escape and cause drowsiness for everyone in the OR. Anesthetic gas is collected by a tank, known as a chemical absorber bed. The fluorine and chlorine in the gas bind to sodium and chloride in the absorber bed to create a salt. As more gas flows into the bed, saturation increases. There is currently no indicator of when the tank is full and the gas has reached unsafe levels. KidsOR and PublicInvention are global organizations combating care inequities and design gaps in low-resource settings. Together, our project seeks to create a solution for the medical personnel who face this daunting challenge daily.

Meeting Notes:

- 10-20 surgeries a day
- Select days for each surgery
- OR is in metal shipping container. In typical OR's the room in itself has a waste gas filtering system
- 1 in 10 have gas scavenging systems
- Room is the size of a 18 wheeler truck
- In most systems there is exhaust in floor or ceiling
- Open circuit system
 - close to the exhaust is a suction fan, avoiding negative pressure on the lung when connected to the child
 - Scrubbing the gases is not very important
 - Take the gases out without making a tripping hazard and get it out of the OR
 - If it going outside someone outside could potentially get sick (unimportant detail)
 - Anesthetic gas is not dangerous for the environment
- Dr. Read solution: put a fan next to exhaust, connected to a tube, run the tube on the floor tape it down and it goes out the window, there is a switch
 - Should it be the anesthesiologist who turns this switch?
 - Modularity is important
 - Senior
 - Source of suction (he wanted a fan next to operating room could be anywhere)
 - The tube is important
 - Scavenging system is where we should start
 - US\$1c charity -> PublicInvention
- Share design history files with Dr. Read
- Decided on meeting structure
 - Bi-weekly meetings
 - Email during weeks (3 by 3 notes - strengths, weaknesses)
 - Loves emails
 - Ask questions
- Add files to GitHub preferred
- Wants everything
- Parameters to consider:
 - What should a typical air flow be?
 - When using a fan CPU ventilation could be enough?
 - What typical outflow is?
 - Consider the air intake for a child.
 - How far would the window or the closest to an exit be?
 - Request pictures of the room?
 - Sivoflourine used typically for anesthetic
 - What level would it affect a person if breathing for 8 hours?
 - Is this detectable with a inexpensive sensor?
- Preventative procedure:
 - Ask KidsOR
 - Typically install a solar panel system
 - Operating independent of the local area
 - Don't have enough power for HVAC unit
 - If solution is a fan out it will also be taking out AC from the room and would cost more money objectively
 - Turn on automatically in the presence of gases
 - Have a test/maintenance mode
 - Can open window turn up fans
 - Ask about ventilation system
- Dr. Stone can provide anesthetic gas for testing

Figure 1: Miro Board with Project Pitch and Meeting Notes with Individual Contributions Circled

We created a project pitch for class that is on our Miro board. This will be helpful later in the DH01 document describing the purpose of the project and creating later project pitches for presentations. I also added a couple questions to ask the KidsOR team when we meet them so we can understand their current solution and how our eventual solution can fit into what they are doing. There is also a list of the meeting notes from the initial team meeting. These notes were transferred onto a word document and combined with the rest of the group, so we are all clear on expectations and what our stakeholder initially told us. All these additions cover the initial meet the stakeholder portion of the design process. Without keeping track of notes and comments from our stakeholder, we cannot start to create user needs.

Theoretical or Numerical Analysis Homework

4. Explain the range of quantitative input values that you would use and the range of values you would expect for the output. Describe how you determined these estimates.

A potential anesthetic gas sensor that could be used for this project is the MPS4SENS Anesthesia Sensor made by Micro-Hybrid [1]. I found this by looking for anesthetic gas sensor datasheets, so I would be able to find an example voltage input value. This device specifically measures CO₂, N₂O, desflurane, and isoflurane. In the specifications, the listed DC output voltage is around 9.35mV. This value can be used as an example quantitative input value for voltage. Since this is a small voltage, a non-inverted amplifier could be used for the Arduino to read the signals. Assuming the Arduino input range is between 0V and 5V, I'm using 2.5V as a voltage output target. In this case, you would calculate $A = V_{out} / V_{in} = 2.5 / 0.00935 = 267$. To calculate the resistor values for a non-inverting amplifier, $A = 1 + R_1/R_2$. The gain needs to be $R_1/R_2 = 266$. Based on this information, we could use a 266KOhm resistor as R₁ and a 1 KOhm resistor as R₂.

5. Bonus (2 pts): Solve the equations using the method or program of your choice and provide an example quantitative result to see if your intuition was correct from the question above.

I plugged in the values below:

$$\begin{aligned} R_1 &= 266k\Omega \\ R_2 &= 1k\Omega \\ V_{in} &= 9.35mV \\ A &= 1 + \frac{R_1}{R_2} \\ &= 1 + \frac{266k\Omega}{1k\Omega} \\ &= 1 + 266 \\ &= 267 \text{ gain} \end{aligned} \quad \begin{aligned} V_{out} &= V_{in} A \\ &= 0.00935 (267) \\ &= 2.50V \end{aligned}$$

Figure 2: Attachment from numerical analysis homework to highlight the equation used and the differences between sensors

When talking about this homework with my group mates we were able to start an interesting conversation about whether we would want to use non-dispersive infrared (NDIR) versus interference filter infrared (IFR) sensors. The solution I found above in this homework was an anesthetic waste gas sensor that was NDIR, while Kaayla based her calculations on a NIR sensor she found online. After doing further research I found that IFR sensors are a subset of NDIR sensors. NDIR sensors are the standard approach to anesthetic waste gas monitoring where a broad-spectrum of infrared light passes through a gas chamber and optical filters sort through these wavelengths before reaching the photodetector [7]. There are also Fourier-Transform Infrared (FTIR) sensors that could be a better solution, and they can capture all different gases at the same time while taking multiple wavelength measurements [7]. The NDIR sensors require multiple measurements for each gas and only takes in separate wavelength bands. The FTIR sensor is also said to calculate faster measurements and is portable, which matches our user needs [7]. This information will be helpful when we are choosing a sensor to build or looking for a sensor to use for our solution. During our stakeholder meeting it was

also brought up that Dr. Bob has a sensor in mind. We have a meeting on Monday morning, so I will bring this information to him and see what his thoughts are.

Design Inputs Map

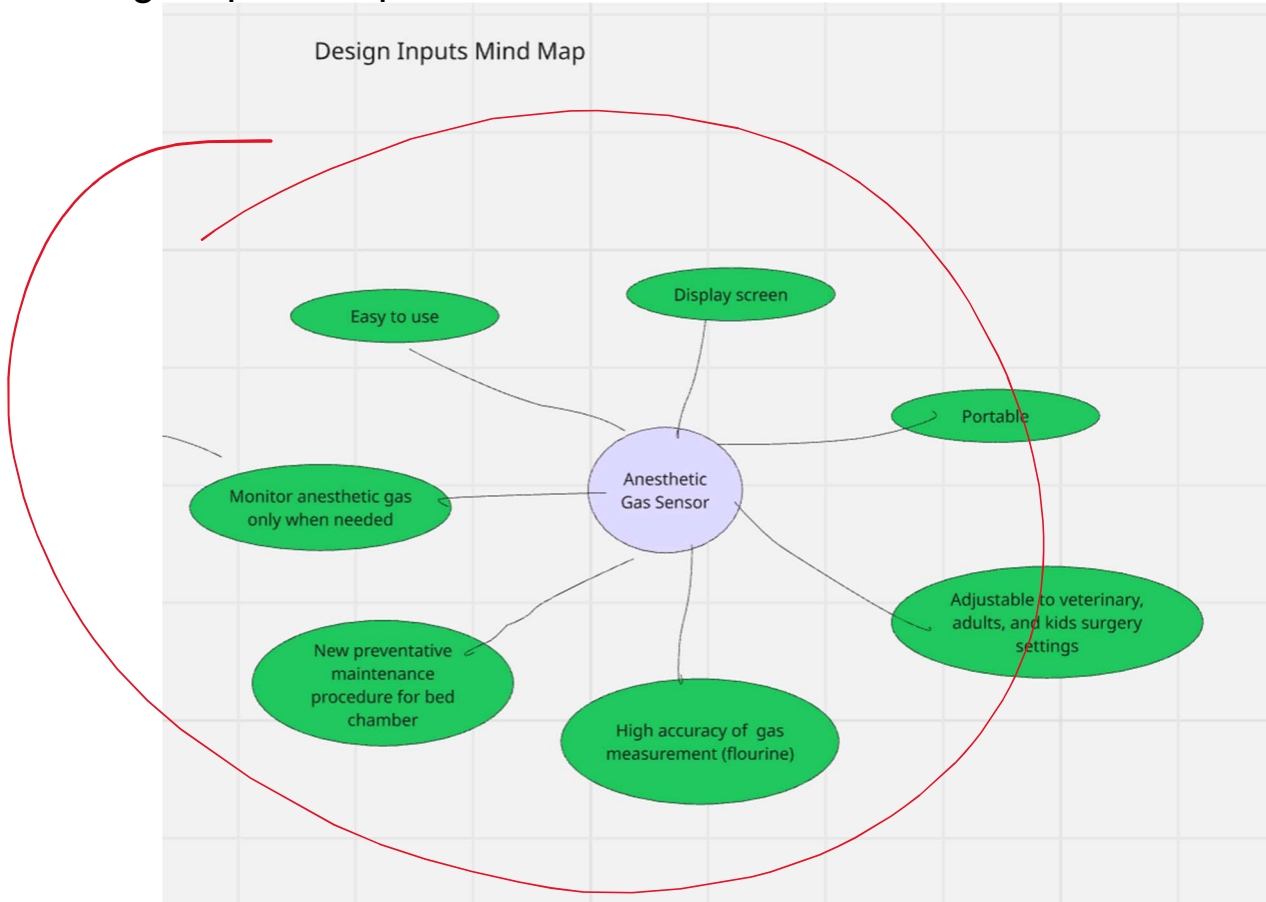


Figure 3: Design Inputs Map with Individual Contributions Circled

I started a design inputs mind map on our group's Miro board, so we can start getting a better picture of the project. Everything in green is a user need, our next step would be to add design inputs attached to the need, so we can further our understanding of the project. Once all members add to the map and some ideas are cut down, it will serve to be useful when brainstorming solutions and coming up with questions for the KidsOR team about their current solution. It will also help the group recenter if we get lost along the way in the design process.

Job Map Update

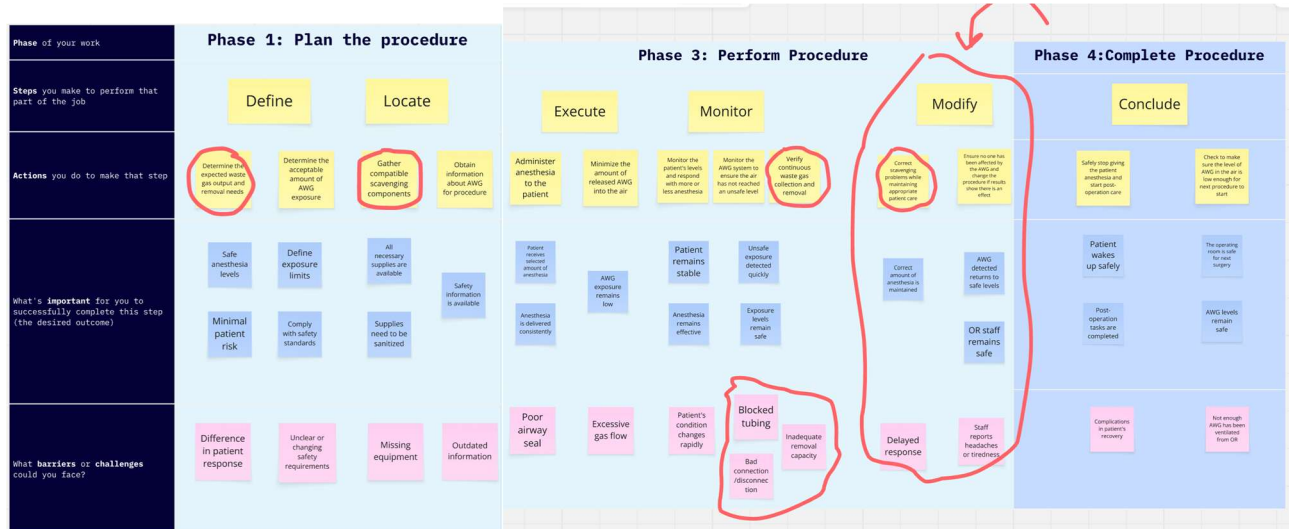


Figure 4: Updated Job Map Since DH01 with Changed Sections Circled

Since the meeting with KidsOR, the project we initially had in mind pivoted. I created an initial job map for DH01 with a sensor system in mind. Our new solution involves creating a standalone active scavenging system. Since our solutions were changed to have a passive design versus a more active design that could directly change the amount of AWG in the air, I also changed the job map a bit to make it align with the new ideas. Phase 2 stayed the same, so it is not pictured above. I also moved “modify” to phase 3 as part of performing a procedure because these adjustments can be made during the procedure not just afterwards. I changed a few challenges under the monitoring section to be specifically catered towards active scavenging systems. By understanding how an anesthesiologist works with the AWG system and how the system works during all phases of a procedure, we can create a solution that fits into the typical schedule of an anesthesiologist and make sure it changes this schedule minimally.

SolidWorks Tutorials

SOLIDWORKS Foundations: Sketching and Extrusion

Course Material

- ☒ Module 1
- ☐ Module 2
- ☐ Module 3

Grades

Notes

Discussion Forums

Messages

Course Info

1 graded assessment left

In this module, you'll learn practical CAD problem-solving techniques in SOLIDWORKS. We'll focus on creating 3D models from 2D sketches using the Extruded Boss/Base feature. This module will also cover essential CAD principles, including sketching and extrusion basics.

Show Learning Objectives

- ☒ Introduction to Practice Basic Sketch and Extrusion (Level 1)
Ungraded Plugin • 15 min
- ☒ Create a simple object (Level 1.1)
Ungraded Plugin • 30 min
- ☒ Create a complex part (Level 1.2)
Ungraded Plugin • 30 min
- ☒ Create a shaft collar (Level 1.3)
Ungraded Plugin • 45 min
- ☒ Create a plastic handle (Level 1.4)
Ungraded Plugin • 45 min

[Resume](#)

☒ Quiz: Practice Basic Sketch and Extrusion
Graded Assignment • 10 min • Grade: -- • Due, Oct 4, 11:59 PM EDT

You're on track with your 7-day learning target this week. Keep up the excellent work!

Edit my weekly learning target

Course timeline

Right on schedule!
You're making great progress! Keep this momentum going and you'll meet your deadlines and estimated end date.

Start date: October 1, 2026

Your next two deadlines

[Quiz: Practice Basic Sketch and Extrusion](#)

Due today by 11:59 PM EDT Graded Assignment

[Quiz: Practice Sketch Tools and End Condition](#)

Due tomorrow by 11:59 PM EDT Graded Assignment

Estimated end date: October 5, 2026

Figure 5: Coursera Dessault Systemes SOLIDWORKS 3D CAD for Education Specialization

After meeting with KidsOR and getting new deliverables for a new solution, I realized that I needed to start working on improving my SolidWorks skills. I found Coursera when trying to find a cheaper way to get a six-sigma yellow belt certification/ specialization, so I could go to engineering expo and have a step up when finding manufacturing internships. I couldn't take the actual SolidWorks class because my organic chemistry lab interferes, but this website acts like a class. It has deliverables to complete every day, quizzes, homework, and discussion posts. By working on my foundational skills, I will be able to create a mockup model for the scavenging system, so we can use it to show stakeholders and our advisors what we have completed. As of right now I am 2 hours into this course and plan to complete this tutorial course by next engineering notebook. The courses include Sketching and Extrusion, Sketching and Reference Geometry, Revolution and Cut Features, Patterns and Sweeps, and Assemblies and Exam Level Part Modeling. In total this full course is 44 hours according to Coursera and sets you up to pass the Certified SolidWorks Associate (CSWA) Exam. I will take this exam realistically in late November or early December.

Anesthetic Waste Gas Equations

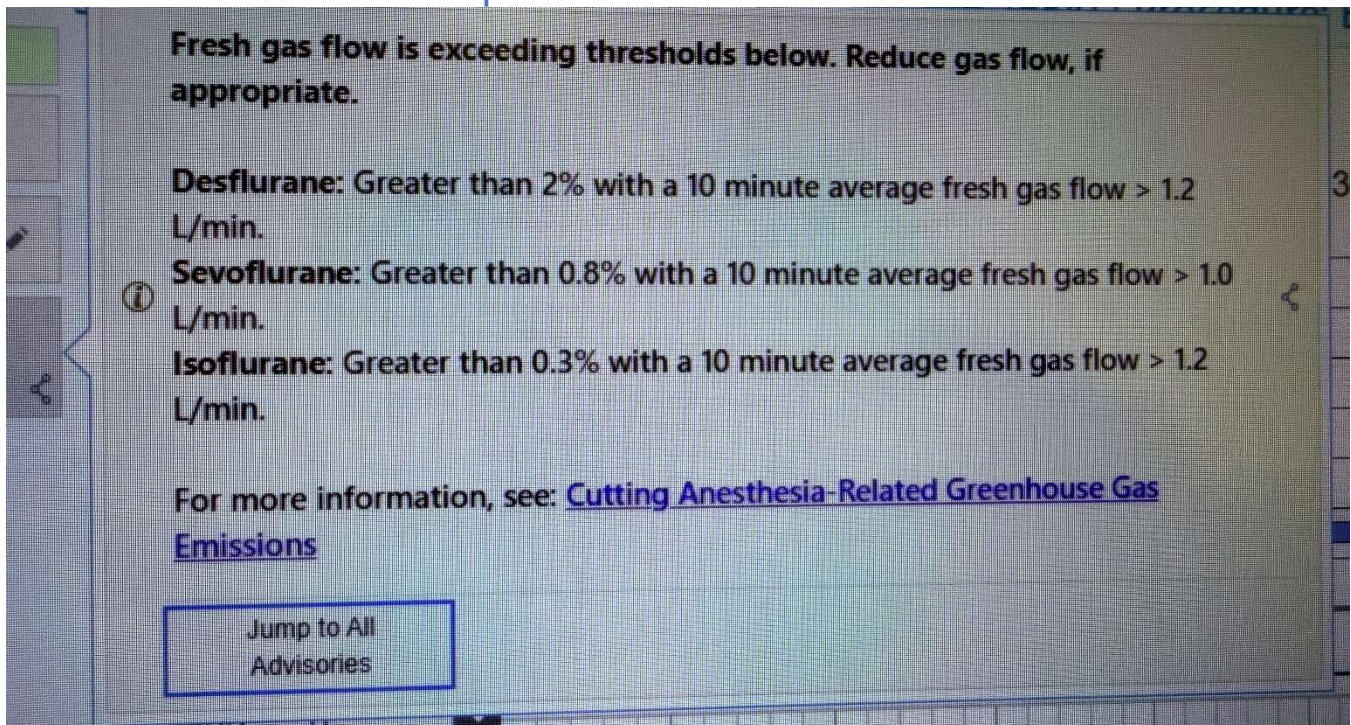


Figure 6: CHOP anesthetic gas low-flow versus high-flow alarm threshold system

The picture attached is directly from my interviewee that works at a hospital. I had asked him if he knew the thresholds for low versus high flow rates when he mentioned that alarm display will show up if there is high flow for a specific period of time. This information can give us an estimate of how much anesthetic gas is typically supplied during surgery. We can use the equation $\text{Gas flow} \times \text{time} = \text{total gas}$. If the surgery is 1L/min for 60 minutes, like literature review 5, there is 60 liters of gas. When using sevoflurane one can multiple $0.8/100 \times 1\text{L} \times 60\text{min} = 0.48\text{ L}$ of sevoflurane vapor. The next step would be to figure out how much of this vapor turns into AWG to then also use this information to help understand how much gas needs to be collected by the scavenging system. This would also help figure

out what type of preventive maintenance procedure we want to create after a specific amount of time the operating room is in use.

References

- [1] K. Samad, Muhammad Saad Yousuf, H. Ullah, Syed Shabbir Ahmed, Khalid Maudood Siddiqui, and A. Latif, "Anesthesia and Its Environmental Impact: Approaches to Minimize Exposure to Anesthetic Gases and Reduce Waste," *Medical Gas Research*, vol. 15, no. 1, Jun. 2024, doi: 10.4103/mgr.medgasres-d-23-00059.
- [2] J. A. Barwise, L. J. Lancaster, D. Michaels, J. E. Pope, and J. M. Berry, "An Initial Evaluation of a Novel Anesthetic Scavenging Interface," *Anesthesia & Analgesia*, vol. 113, no. 5, pp. 1064–1067, Nov. 2011, doi: 10.1213/ane.0b013e31822c9a2c.
- [3] H.-B. Deng, F.-X. Li, Y.-H. Cai, and S.-Y. Xu, "Waste Anesthetic Gas Exposure and Strategies for Solution," *Journal of Anesthesia*, vol. 32, no. 2, pp. 269–282, Feb. 2018, doi: 10.1007/s00540-018-2448-1.
- [4] J. C. Nesbitt, D. A. Krageschmidt, and M. C. Blanco, "A Novel Approach to Scavenging Anesthetic Gases in Rodent Surgery," *Journal of Occupational and Environmental Hygiene*, vol. 10, no. 9, pp. D125-D131, Sep. 2013, doi: [10.1080/15459624.2013](https://doi.org/10.1080/15459624.2013).
- [5] Y. Shiraishi and K. Ikeda, "Uptake and Biotransformation of Sevoflurane in Humans: A Comparative Study of Sevoflurane With Halothane, Enflurane, and Isoflurane," *Journal of Clinical Anesthesia*, vol. 2, no. 6, pp. 381–386, Nov. 1990, doi: [10.1016/0952-8180\(90\)90024-W](https://doi.org/10.1016/0952-8180(90)90024-W).
- [6] N. A. Shallik, W. A. Eldegowa, M. O. Zekri, and M. F. Alhammad, "Scavenging, Medical Gas Pipelines, and Vacuum Systems," *Improving Anesthesia Technical Staff's Skills*, pp. 137–150, 2022, doi: [10.1007/978-3-030-88849-7_10](https://doi.org/10.1007/978-3-030-88849-7_10).
- [7] "Anesthetic Waste Gas Monitoring," Gasmet.com. Accessed: Sep. 11, 2026. [Online]. Available: <https://www.gasmet.com/applications/safety/anesthetic-waste-gas-monitoring/>